

BHASHASETU

भाषा सेतु

AI-Powered Vernacular Pedagogy and Real-Time Translation Tool for Mother Tongue-Based Primary Education

Problem Statement ID	26042
Organization	Government of Jharkhand — Department of Higher & Technical Education
Theme	Smart Education
Target Languages	Ho, Mundari, Santhali (Ol Chiki/Devanagari), Hindi
Framework	NIPUN Bharat & Vidya Pravesh — Foundational Literacy & Numeracy (FLN)
Target Device	Low-cost Android tablets, Android 9+, ≥2 GB RAM

GitHub: github.com/vard143/SMART_INDIA_HACKATHON

1. Executive Summary

BhashaSetu is an AI-powered vernacular pedagogy and real-time translation platform designed for mother-tongue-based primary education. It bridges communication between Hindi-speaking teachers and learners using Jharkhand's tribal languages, while combining translation, curriculum support, storybooks, worksheets, phonics practice, and offline-first operation on low-cost Android devices.

The project is aligned with foundational literacy and numeracy (FLN) requirements and is designed around NIPUN Bharat and Vidya Pravesh-style primary education workflows.

2. Problem Statement

Primary classrooms can face a language gap when teachers and learners do not share the same mother tongue. This can affect classroom interaction, instruction delivery, pronunciation practice, and access to learning materials.

BhashaSetu addresses this gap through a classroom voice bridge, vernacular learning resources, and an offline-first architecture intended for environments with limited or intermittent internet access.

3. Proposed Solution

- Voice-to-Voice Bridge: teacher Hindi → tribal-language support and student tribal-language → Hindi support.
- NIPUN FLN Curriculum: lesson scripts, teacher prompts, and learning content.
- Worksheet & Flashcard Studio: bilingual printable and interactive learning materials.
- Bolo Aur Jeeto: pronunciation practice with phonemic scoring, stars, and streaks.

4. Major Features

Feature	Description
Real-Time Classroom Voice Bridge	Voice-to-voice classroom assistance, teleprompter-style support, audio playback, and one-tap classroom action chips.
NIPUN Bharat FLN Curriculum	Balvatika and Class 1–3 lesson scripts, teacher prompts, and curriculum-oriented workflows.
Vernacular Storybook Hub	Bilingual storybook reading and AI-assisted vernacular lesson/story content.
Worksheet & Flashcard Studio	Printable A4 worksheets and flashcards for matching, counting, tracing, fill-blanks, and phonics.
Bolo Aur Jeeto	Pronunciation/phonics practice with AI-assisted phonemic scoring, stars, and streaks.
Tribal Cultural Companion	Cultural and pedagogical guidance covering festivals and classroom do's/don'ts.
Offline-First Edge Runtime	Local phrase matching, embedded language resources, browser-side caching, and local audio support.

5. Curriculum & Pedagogy

NIPUN Bharat / FLN Support

The curriculum module is structured around foundational literacy and numeracy workflows, providing lesson scripts, teacher prompts, and vernacular support for primary classroom activities.

Vernacular Storybook Hub

The storybook area provides bilingual reading experiences and supports vernacular lesson generation so learning content can be presented in a form accessible to mother-tongue learners.

Worksheet & Flashcard Studio

The studio is designed for printable A4 learning materials and interactive flashcards, including picture-to-word matching, FLN counting, letter tracing, fill-in-the-blank activities, and phonics exercises.

6. Offline-First Architecture

- PWA Layer: responsive classroom application for low-cost Android tablets and school laptops.
- Local Storage: IndexedDB for client-side data and Service Worker caching for offline assets.
- Language Engine: hybrid fast rule/fuzzy phrase matching with a tribal morphological parser.
- Audio Layer: Web Audio acoustic phoneme synthesizer with speech-to-text fallback.
- Backend API: FastAPI/Python endpoints for translation, curriculum generation, and synchronization.

Backend API Endpoints

- /api/translate
- /api/curriculum/generate
- /api/sync/bundle

7. Technology Stack

Layer	Technology / Approach
Frontend	Progressive Web App (PWA), responsive web interface
Backend	Python + FastAPI
Offline Data	IndexedDB + Service Worker cache
Language Processing	Rule/fuzzy phrase matcher + tribal morphological parser
Audio	Web Audio acoustic phoneme synthesizer + STT fallback
Learning Content	JCERT / NIPUN-oriented curriculum and vernacular content
Documents	A4 vector worksheet / flashcard generation

8. Repository Structure

- backend/ — backend API and server-side implementation.
- frontend/ — PWA/frontend implementation.
- bundles/ — packaged learning/language resources.
- data/ — project data resources.
- ppt_assets/ — presentation assets.
- scripts/ — project utility/build scripts.
- tests/ — project tests.
- Documentation — README, SIH idea submission, full project documentation, and QA/production verification report.

9. Target Hardware & Deployment

- Low-cost Android tablets
- Android 9+
- At least 2 GB RAM
- School laptops / responsive browser environments
- Offline-first classroom environments with optional cloud synchronization

10. End-to-End Classroom Workflow

- 1. Teacher selects a lesson or starts the classroom voice bridge.
- 2. Teacher speaks or selects a classroom action in Hindi.
- 3. The language pipeline identifies a phrase and maps it to the supported tribal-language representation.
- 4. The student receives vernacular audio/text support.
- 5. Student response can be captured and bridged back toward Hindi.
- 6. The teacher continues using curriculum prompts, storybooks, worksheets, or flashcards.
- 7. Pronunciation activities can be practiced in the Bolo Aur Jeeto arena.
- 8. When connectivity is unavailable, the offline runtime continues using locally available resources.

11. Documented Performance & Verification

The following values are reported in the project repository's documentation and verification material. They are repository-reported figures, not an independent benchmark performed for this document.

Metric	Repository-reported value
Voice translation latency	0.4–1.2 seconds
Target latency	<3 seconds
Offline operation	Zero-internet operation
RAM footprint	Approximately 135 MB
Language coverage	Ho, Mundari, Santhali
Worksheet format	A4 vector PDF with Ol Chiki & Devanagari support

12. Target Users

- Primary-school teachers in multilingual classrooms
- Students learning through their mother tongue
- Schools using low-cost Android tablets or laptops
- Education departments and institutional FLN programs
- Curriculum and teacher-support teams

13. Innovation Areas

- Combines classroom translation and pedagogy rather than treating translation as a standalone feature.
- Offline-first design keeps core classroom functions available without continuous internet.
- Supports vernacular language resources alongside curriculum and teacher prompts.
- Integrates worksheets, storybooks, flashcards, and pronunciation practice into one classroom workflow.
- Targets low-cost hardware and practical school deployment.

14. Advantages

- Reduces the classroom communication gap between teacher and learner.
- Supports mother-tongue-based learning workflows.
- Provides reusable bilingual teaching resources.
- Designed for constrained connectivity and low-cost devices.
- Combines translation, learning content, practice, and classroom support in one platform.

15. Future Expansion

- Expansion to additional Indian tribal and regional languages.
- Broader state-wise curriculum bundles and multi-state deployment.
- Additional offline language models and speech resources.
- Teacher analytics and learning-progress dashboards.
- Expanded synchronization and institutional administration capabilities.

16. Project Impact

BhashaSetu is intended to make primary education more accessible in multilingual classrooms by combining vernacular communication with foundational learning resources. Its offline-first architecture is aimed at practical deployment in schools where connectivity cannot be assumed.

17. Technical Quick Reference

Item	Details
Project	BhashaSetu
PS ID	26042
Theme	Smart Education
Backend	FastAPI / Python
Frontend	PWA
Core languages	Hindi, Ho, Mundari, Santhali
Offline storage	IndexedDB
Caching	Service Worker
Core APIs	translate, curriculum/generate, sync/bundle
Target	Low-cost Android tablets / school laptops

18. One-Minute Explanation for Judges

BhashaSetu is an AI-powered vernacular education platform for multilingual primary classrooms. It bridges Hindi-speaking teachers and tribal-language learners through a real-time classroom voice bridge, while also providing NIPUN Bharat-oriented FLN content, bilingual storybooks, printable worksheets, flashcards, and pronunciation practice. The platform is built as an offline-first PWA for low-cost Android devices, so core learning and language support can continue even when internet connectivity is unavailable.

19. Project Repository

GitHub: https://github.com/var143/SMART_INDIA_HACKATHON

The repository contains the implementation, presentation assets, project documentation, scripts, tests, and verification material used for the SIH project.